Software Development Project — G&G City · Lab Backlog Sprint 1

LAB S3: Home Page + Publishing — Ayman Al-Zahrani

Topic: Integrating Components into One Game Loop + Deploying to GitHub Pages

Learning outcomes

By the end of this lab, students will be able to:

- Connect independent modules through their public functions only
- Write an async game loop with `await`
- Save progress safely in `localStorage` and publish with GitHub Pages

🎯 Problem Scenario

You are the meeting point of all teams. You do not rewrite their code: you call it in the right order.

- **From C1/C2: `createCat` · from C3: `createBox`, `createCommandBox` · from S2: `createScenes`**
- **From P1: `data/lessons.json`. Until a part is ready, test with the files in its folder**

Your files: `index.html`

Required Outputs

Build:

1. **Load the three scripts and `lessons.json`**
2. **Flatten all lessons of all stations into one ordered list of steps**
3. **`showStep(i)`: walk to the station if needed, place the cat and box, show the intro, create the command box**
4. **On a result: charge → spell → say; on success update the box, save progress, go to the next step**
5. **Save / load progress, and publish on GitHub Pages**

Code is given only for the hard parts. Parts marked 📝 **Note** are easy: follow the note and write them yourself. Test with **Live Server** in VS Code (`fetch` does not work when you double-click a file).

Model Answer:

Task 1: Page layout + scripts

> 📝 **Note (no code needed):** Normal `<script src="...">` tags for `scenes/scenes.js`, `cat/cat.js`, `box/box.js` at the end of `<body>`. Two containers: `#screen` and `#command`. Size the characters inside the stage with `.gg-stage .gg-cat { width: 17% }` and `.gg-stage .gg-box { width: 15% }`. Style `.gg-command` like a dark terminal.

Task 2–5: Progress + the game loop (hard part)

`append(box.el, cat.el)` **moves** the existing elements into the new station: we never create a second cat. The `try/catch` stops the game from crashing in private browsing where storage is blocked.

```
const SAVE_KEY = 'gg-city-progress';
function loadProgress() { try { return Number(localStorage.getItem(SAVE_KEY)) || 0; } catch
{ return 0; } }
function saveProgress(n) { try { localStorage.setItem(SAVE_KEY, String(n)); } catch {} }

// ---------------------------------------------------------------------
// Hard part 2: the game loop — one step = one lesson
// ---------------------------------------------------------------------
async function start() {
  const data = await fetch('data/lessons.json').then(r => r.json());
  const stations = [...data.stations].sort((a, b) => a.order - b.order);

  // Flatten: [{ station: 0, lesson }, { station: 0, lesson }, { station: 1, lesson } ...]
  const steps = stations.flatMap((st, s) =>
    [...st.lessons].sort((a, b) => a.order - b.order).map(lesson => ({ station: s, lesson
})));

  const scenes  = createScenes(document.getElementById('screen'), stations);
  const cmdHost = document.getElementById('command');
  const cat = createCat(scenes.stage(0));
  const box = createBox(scenes.stage(0));
  await Promise.all([cat.ready, box.ready]);

  async function showStep(i) {
    if (i >= steps.length) { cmdHost.innerHTML = ''; cat.say('0 ,'🎉 !أنهيت كل المحطات');
return; }
    const { station, lesson } = steps[i];

    if (scenes.current() !== station) {            // walk to the next station
      cat.walk();
      await scenes.goTo(station);
      cat.idle();
    }
```

```
    scenes.stage(station).append(box.el, cat.el);   // move both characters into this
station

    box.setState(lesson.boxStateBefore);
    cat.say(lesson.cat.intro + '\n' + lesson.cat.task, 0);

    cmdHost.innerHTML = '';
    createCommandBox(cmdHost, lesson, async result => {
      await cat.charge();
      await cat.spell(result.ok);
      cat.say(result.feedback, 0);
      if (!result.ok) return;

      box.setState(lesson.boxStateAfter);
      if (lesson.boxLabel) box.setLabel(lesson.boxLabel);
      saveProgress(i + 1);
      setTimeout(() => showStep(i + 1), 2500);      // let the learner read the success
line
    }, data.matching).focus();
  }

  showStep(loadProgress());
```

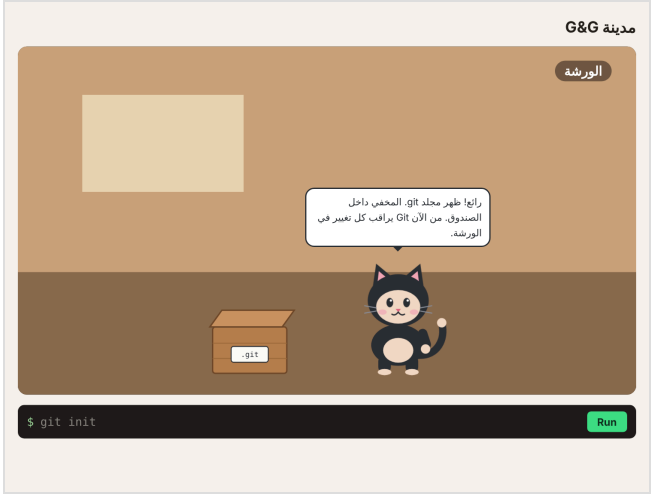
Task 5: Publish

> 📝 **Note (no code needed):** On GitHub: **Settings → Pages → Source: Deploy from a branch → main / (root) → Save**. After about a minute the site is at `https://mr-maxvx.github.io/G_G-City/`. Open it on a phone too. Add the link at the top of `README.md`.

Expected Output



First visit: the cat explains the task



git init → success, next step in 2.5 s

Submission

```
git pull
git add index.html
git commit -m "ربط المشاهد والشخصيات في الصفحة الرئيسية"
git push
```